SANTA CLARA UNIVERSITY
Leavey School of Business

Natural Language Processing

ISBA 2411 | Summer 2026

Santa Clara University | *Leavey School of Business*

Course Number	ISBA 2411
Course Title	Natural Language Processing
Credits	4 Units (20 classes, 10 weeks, ~32 hours, 1 hr 35 min/class)
Term	Summer 2026
Class Hours	Tuesday & Thursday, 5:45 – 7:20 PM
Instructor	Professor Malin
Instructor URL	github.com/JulesMalin/isba2411-nlp
Office Hours	Tuesday, 7:30 – 8:30 PM
Office Location	216UU, Lucas Hall
Platform	Google Colab (included in SCU Google Suite); SCU WAVE cluster available for HPC (VPN required off-campus)
Prerequisites	Python programming required (see related MSBA/MSIS courses); R is useful but not required. Prior ML coursework is not required but helpful. A free GitHub Student account is required; install and learn GitHub Desktop before Week 1. A Google account with access to Google Colab is required (included in your SCU Google Suite); confirm you can open and run a Colab notebook before Week 1.

1. Course Overview

This course provides a rigorous, hands-on foundation in Natural Language Processing and Large Language Models, tracing the arc from classical linguistic structure to production-grade generative AI systems. Drawing on Jurafsky & Martin's Speech and Language Processing (J&M), students build core competencies in tokenization, POS tagging, named entity recognition, text classification, and language modeling, grounded in the probabilistic and neural frameworks that underpin modern NLP. Business applications woven throughout include sentiment analysis, document similarity, information extraction, and economic text analysis (e.g., FOMC transcripts and financial news).

The course then advances into transformer-based models and large language models using Alammar & Grootendorst's Hands-On Large Language Models (HOLLM) and Tunstall et al.'s NLP with Transformers (Tunstall). Students implement BERT, RoBERTa, and generative LLMs via the Hugging Face ecosystem, apply prompt engineering and parameter-efficient fine-tuning (LoRA/QLoRA), and build retrieval-augmented generation (RAG) pipelines with LangChain and LangGraph. The final arc covers responsible AI (evaluation benchmarks, fairness, RLHF, red-teaming) and LLMOps: monitoring, governance, deployment architectures, and build-vs-buy decisions.

All course code and materials run on Google Colab (already included in your SCU Google Suite). We will also use jupyter-ai, a modern extension to Jupyter Lab/Notebooks. The SCU WAVE cluster is available for large-scale, high-performance computing (off-campus use requires a VPN).

2. Learning Objectives

Upon successful completion of this course, students will be able to:

- Explain the NLP task landscape, spanning tokenization and parsing to generation and agentic workflows, and situate each within organizational and governance contexts
- Apply subword tokenization, POS tagging, NER, and constituency parsing to raw text using spaCy and Hugging Face, and analyze how linguistic structure shapes downstream model behavior
- Build and evaluate supervised text classification pipelines using GLMNet, FastText, spaCy, and AutoGluon, interpreting term-document matrices and n-gram language models
- Implement text retrieval, sentiment analysis, and document similarity pipelines using PyTorch and Hugging Face, connecting deep learning fundamentals to practical NLP workflows
- Apply word embeddings and topic modeling techniques, including t-SNE visualization and dimension reduction, applied to real-world corpora such as economic news and FOMC transcripts
- Explain the transformer architecture and interpret attention patterns in encoder and decoder models, and implement BERT, RoBERTa, and Hugging Face pre-training workflows
- Design and implement retrieval-augmented generation (RAG) systems using LangChain and LangGraph, incorporating multimodal LMs, agentic workflows, and hallucination mitigation strategies
- Evaluate NLP systems using benchmark frameworks and structured red-teaming, applying RLHF concepts, Shapley values, and fairness metrics to assess bias, robustness, and safety
- Design production NLP architectures addressing LLMOps, monitoring and drift detection, governance, privacy, and build-vs-buy deployment tradeoffs
- Implement end-to-end NLP task pipelines (NER, extractive QA, text classification) using Hugging Face, with multi-metric evaluation, error analysis, and weak supervision strategies
- Communicate NLP system value, risk, and ROI to non-technical stakeholders through data-driven storytelling, executive framing, and governance disclosure

3. Required Texts & Materials

Three texts are used throughout the course and referenced in weekly readings:

J&M Jurafsky, D. & Martin, J.H. (2026). Speech and Language Processing (3rd ed. draft, released Jan 6, 2026). Free online: web.stanford.edu/~jurafsky/slp3/

HOLLM Alammar, J. & Grootendorst, M. (2024). Hands-On Large Language Models. O'Reilly Media. ISBN: 978-1098150969

Tunstall Tunstall, L., von Werra, L., & Wolf, T. (2022). Natural Language Processing with Transformers (Revised ed.). O'Reilly Media. ISBN: 978-1098136796

Students will need access to Python 3.10+ and Google Colab or an equivalent GPU environment. A free GitHub Student account is required; install and learn GitHub Desktop before Week 1. A Google account with access to Google Colab (included in your SCU Google Suite) is required; confirm Colab access before Week 1.

4. Grading & Assessment

Component	Weight	Description
Final Team Project Milestones (Assignments 1–4)	30%	Interim milestone deliverables submitted throughout the course: Business Brief, Project Baseline & Representation, Model Adaptation Experiment, and System Prototype. All milestones are team deliverables. Collaboration within a team is expected and encouraged; collaboration across teams is not permitted.
Midterm Exam	30%	Individual, in-class exam (Week 6, Class 12). Covers Weeks 1–5 content. You may use Gen AI tools provided you cite their use.
Final Team Project: System & Presentation (Assignment 5)	30%	Production-ready team system with full documentation, evaluation results, and a governance/risk appendix, presented live across both Week 10 classes (Aug 18 and Aug 20) (max 4 students; no exceptions). Evaluated against the Team Project Rubric (separate companion document).
In-Class Final Team Project Updates & Participation	10%	Teams give brief in-class updates on their Final Team Project during the weeks between milestones (Weeks 2, 4, 6, and 8) and an in-class demo during final presentations (Aug 20). This component grades these in-class updates and the demo plus overall participation and attendance throughout the term.

The detailed Team Project Rubric — five weighted dimensions scored 1–5 — is provided in the companion document, “Team Project Rubric.”

Grading Scale

A (95–100)	**A- (90–94)**	**B+ (87–89)**	**B (83–86)**	**B- (80–82)**	**C+ (77–79)**	**C (73–76)**	**C- (70–72)**	**F (<70)**

5. Weekly Schedule

Class meets Tuesday and Thursday, 5:45–7:20 PM. The table below summarizes the full course schedule. Detailed weekly learning outcomes are provided in the companion document, "Weekly Learning Outcomes."

Wk	Theme	Key Topics	Readings	Assignment
1 Jun 16, 18	NLP as a Socio-Technical System	Ambiguity; task landscape; tokenization; LMs vs LLMs; governance; jupyter-ai intro	J&M Ch. 2–3; HOLLM Ch. 1; Tunstall Ch. 1	Assignment 1 Milestone 1: Business Brief (due Sun, Jun 21)
2 Jun 23, 25	Language as Structured Data	Subword tokenization; POS; NER; parsing; embeddings; multilingual pipelines; ML review; linear algebra	J&M Ch. 17–18; HOLLM Ch. 2; Tunstall Ch. 4	Team Update In Class
3 Jun 30, Jul 2	Supervised Learning for NLP	POS tagging; term-document matrix; text classification (GLMNet, FastText, spaCy); AutoGluon multimodal	J&M Ch. 4; HOLLM Ch. 4; Tunstall Ch. 2	Assignment 2 Milestone 2: Baseline & Representation (due Sun, Jul 12)
4 Jul 7, 9	Text Retrieval & Deep Learning Intro	Web scraping; sentiment; summarization; doc similarity; deep learning intro; PyTorch	J&M Ch. 6; HOLLM Ch. 8; Tunstall Ch. 6	Team Update In Class
5 Jul 14, 16	Deep Learning & Topic Modeling	Deep learning (continued); Hugging Face sentiment & QA; topic modeling; dimension reduction	J&M Ch. 8; HOLLM Ch. 5; Tunstall Ch. 3	Assignment 3 Milestone 3: Model Adaptation (due Sun, Jul 26)
6 Jul 21, 23	Embeddings & Midterm	Word embeddings; t-SNE; economic news analysis; FOMC transcripts | Midterm (Class 12)	J&M Ch. 5; HOLLM Ch. 10	Team Update In Class (Jul 21) Midterm (Class 12, Jul 23)
7 Jul 28, 30	Language Models & Transformers	Generalized LMs; Transformers paper; BERT; RoBERTa; Hugging Face; Gluon BERT pre-training	J&M Ch. 7, 10; HOLLM Ch. 3; Tunstall Ch. 10	Assignment 4 Milestone 4: System Prototype (due Sun, Aug 9)
8 Aug 4, 6	LMs, Generative AI & RAG	Multimodal LMs; NLG; LangChain; RAG; LangGraph; Agentic workflows; Streamlit/Shiny	J&M Ch. 11; HOLLM Ch. 7, 9; Tunstall Ch. 5, 7	Team Update In Class
9 Aug 11, 13	Evaluation, Ethics & Responsible AI	RLHF; reinforcement learning; Shapley values; privacy; fairness in AI/ML; evaluating LLMs	J&M Ch. 9; Tunstall Ch. 9	Assignment 5 Milestone 5: Final System & Demo (due Sun, Aug 23)
10 Aug 18, 20	Final Presentations & Demos	Team final presentations & demos across both classes (Aug 18 & 20); LLMOps & production deployment; model efficiency in production; future directions & trends; course wrap-up	HOLLM Ch. 11–12; Tunstall Ch. 8, 11	Final Presentations (Aug 18 & 20)

6. Academic Policies & Institutional Notices

Academic Integrity

The Academic Integrity pledge is an expression of the University's commitment to fostering a culture of integrity at Santa Clara University. All students are expected to abide by the SCU Academic Integrity Policy.

You are welcome to use Gen AI tools like LLMs for coursework and in exams, provided you make a note when you do. It would be paradoxical not to allow use of LLMs in a class on AI. However, it is important for the instructor to see how well you use these tools; please cite all tool use when you avail of it.

Course sharing sites such as Chegg are prohibited. Collaboration within a homework group is expected and encouraged, but collaboration across homework groups is not permitted. A student guilty of a dishonest act may receive a grade of F for the course.

AI Tools Policy

This course covers AI systems: you are expected to engage with them critically, not merely use them passively. Specific rules:

- All AI-generated content must be clearly disclosed with a brief explanation of how and why the tool was used
- You are responsible for the correctness, integrity, and originality of all submitted work
- AI tools may be used during the Midterm exam, provided use is cited
- Using AI to complete an assignment without engagement or disclosure constitutes academic dishonesty

Late Work

Assignments submitted late without prior approval will receive a 10% deduction per day, up to a maximum of 50%. Assignments more than 5 days late will receive zero. If you anticipate difficulty meeting a deadline, contact the instructor at least 48 hours in advance. Final Team Project milestones are group deliverables and extensions are rarely granted.

Accessibility & Accommodations (OAE)

If you have a documented disability for which accommodations may be required, contact the Office of Accessible Education (oae@scu.edu | scu.edu/oae) as soon as possible. If you have already arranged accommodations through OAE, request them via your MyOAE portal and discuss them with the instructor within the first two weeks of class. At least two weeks' notice is recommended for proctored exams or other accommodations.

Title IX & Discrimination

Santa Clara University is committed to providing a safe learning environment free of all forms of discrimination, sexual harassment, and sexual violence. California law (SB 493) requires the instructor to report information about sexual harassment or misconduct to the SCU Equal Opportunity and Title IX Office at (408) 551-3043. Confidential support is available through the SCU Wellness Center and CAPS.

Pregnant & Parenting Students

SCU does not discriminate against any student on the basis of pregnancy or related medical conditions. Absences due to pregnancy-related conditions will be excused as medically necessary, with opportunity to make up missed work. Contact the Office of Accessible Education (OAE) or the Title IX Office for accommodations support.

Attendance & Participation

Regular attendance is expected and is assessed through the In-Class Final Team Project Updates & Participation component (10% of the course grade; see Grading & Assessment). Teams give a brief in-class project update during Weeks 2, 4, 6, and 8 and an in-class demo during final presentations (Aug 20), and attendance is taken at these sessions. More than two unexcused absences may negatively affect your participation grade. If you must miss class, you are responsible for obtaining notes and completing any in-class activities; please notify the instructor in advance when possible.

Classroom Recordings

All online class meetings will be recorded and made available on Camino. Dissemination or sharing of any classroom recording without the permission of the instructor is prohibited and may result in disciplinary action. At the instructor's discretion, violations may also have an adverse effect on the student's grade.

Technology Support

For Camino support, contact caminosupport@scu.edu or call 408-551-3572. For Zoom assistance, contact Media Services at mediaservices@scu.edu or 408-554-4520. For SCU network and computing support, contact the Technology Help Desk at techdesk@scu.edu or 408-554-5700.

Wellness

Your wellbeing matters. Jesuit education is grounded in cura personalis, concern for the whole person. Attend to your sleep, nutrition, and mental health. SCU provides free counseling through CAPS (408-554-5220, 24/7 support line) and additional resources through the Wellness Center (scu.edu/wellness) and Culture of Care (scu.edu/osl/culture-of-care). Please reach out early, to the instructor or to SCU resources, whenever you need support.

Copyright

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This syllabus is subject to change. Any modifications will be announced via Camino with advance notice.